

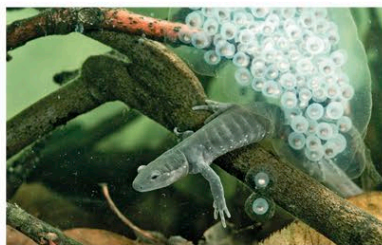
AMPHIBIANS

Amphibians are cold-blooded vertebrates that thrive in freshwater habitats. Some spend their whole lives in water; others only require water to breed. On land, they must find damp places as their skin is permeable and does not protect them from drying out.

PHYLUM	CHORDATA
CLASS	AMPHIBIA
ORDERS	3
FAMILIES	74
SPECIES	8,212



Large numbers of male and female Costa Rican golden toads gather in a pool to mate. The species is now extinct.



A female of the North American Jefferson's salamander attaches a mass of eggs to a submerged twig in early spring.



When the eggs hatch, the male red-backed poison frog carries the tadpoles to bromeliad pools in the canopy.

DEBATE

OUR BIGGEST CHALLENGE?

A third of the world's amphibians face extinction in the near future and this poses a huge conservation challenge. This danger of extinction is largely due to the destruction and pollution of freshwater habitats, but amphibians are also threatened by the global spread of the disease chytridiomycosis, caused by a fungus that invades their soft skin.

The three orders of living amphibians are thought to share a common ancestor, but their origins are still uncertain. There is an enormous gap in the fossil record between the first land animals to evolve from fish – the tetrapods, some 375 million years ago – and a frog-like creature that lived 230 million years ago.

Amphibians have a unique and complex life cycle and occupy very different ecological niches at different stages of their lives. For most amphibians, their eggs hatch into larvae – called tadpoles in frogs and toads – that live in water, often in very high densities, and feed mostly on algae and other plant material. As larvae, they grow rapidly and then undergo a complete change of form – metamorphosis – to become land-living adults. As adults, all amphibians are carnivores, most feeding on insects and other small invertebrates. They typically lead secretive and solitary lives, except when they return to ponds and streams to breed. Most amphibians thus require two very different habitats during their lives: water and land. During metamorphosis, they undergo a wide variety of anatomical and physiological changes – from aquatic creatures that swim with a tail and breathe through gills, to terrestrial animals that move with four limbs and breathe through lungs.

DIVERSE PARENTAL CARE

Some amphibians produce huge numbers of eggs that are left to fend for themselves, so that only a very few will survive. Many others have evolved various forms of parental care. This is generally associated with the production of far fewer offspring, so that reproductive success is achieved by caring for a manageable number of young, rather than by producing as many eggs as possible. Parental care takes many forms, including defending eggs or larvae against predators, feeding tadpoles with unfertilized eggs, and carrying tadpoles from one place to another. In some species, such as the midwife toads and some poison frogs, parental care is the duty of the father. In many salamanders and caecilians, the mother is the sole protector of the young. In a few frog species, mother and father establish a durable pair bond and share parental duties.

READY TO HATCH >

Tadpoles of the Tanzanian Mitchell's reed frog wriggle within their egg membranes, just prior to hatching.

